

SUV

Worcester

2024-01-24

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.4.4      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tidymodels)
```

```
## -- Attaching packages ----- tidymodels 1.1.1 --
## v broom       1.0.5      v rsample    1.2.0
## v dials       1.2.0      v tune       1.1.2
## v infer       1.0.5      v workflows  1.1.3
## v modeldata   1.2.0      v workflowsets 1.0.1
## v parsnip     1.1.1      v yardstick  1.2.0
## v recipes     1.0.9
## -- Conflicts ----- tidymodels_conflicts() --
## x scales::discard() masks purrr::discard()
## x dplyr::filter()   masks stats::filter()
## x recipes::fixed() masks stringr::fixed()
## x dplyr::lag()      masks stats::lag()
## x yardstick::spec() masks readr::spec()
## x recipes::step()   masks stats::step()
## * Search for functions across packages at https://www.tidymodels.org/find/
```

```
library(ggforce)
```

```
library(mctest)
```

```
library(olsrr)
```

```
##
## Attaching package: 'olsrr'
##
## The following object is masked from 'package:datasets':
##
##     rivers
```

```
library(jtools)
```

```
##  
## Attaching package: 'jtools'  
##  
## The following object is masked from 'package:yardstick':  
##  
##   get_weights
```

```
library(ggcorrplot)  
library(yardstick)  
library(car)
```

```
## Loading required package: carData  
##  
## Attaching package: 'car'  
##  
## The following object is masked from 'package:dplyr':  
##  
##   recode  
##  
## The following object is masked from 'package:purrr':  
##  
##   some
```

```
library(moments)  
library(GGally)
```

```
## Registered S3 method overwritten by 'GGally':  
##   method from  
##   +.gg   ggplot2
```

```
library(psych)
```

```
##  
## Attaching package: 'psych'  
##  
## The following object is masked from 'package:car':  
##  
##   logit  
##  
## The following objects are masked from 'package:scales':  
##  
##   alpha, rescale  
##  
## The following objects are masked from 'package:ggplot2':  
##  
##   %+%, alpha
```

```
library(fastDummies)
```

```
## Thank you for using fastDummies!  
## To acknowledge our work, please cite the package:  
## Kaplan, J. & Schlegel, B. (2023). fastDummies: Fast Creation of Dummy (Binary) Columns and Rows from
```

```
library(lmtest)
```

```
## Loading required package: zoo  
##  
## Attaching package: 'zoo'  
##  
## The following objects are masked from 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
SUV <- read_csv("SUV.csv") %>% as_tibble()
```

```
## Rows: 400 Columns: 4  
## -- Column specification -----  
## Delimiter: ","  
## dbl (4): GEN, AGE, SAL, SUV  
##  
## i Use 'spec()' to retrieve the full column specification for this data.  
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
summary(SUV)
```

	GEN	AGE	SAL	SUV
## Min.	:0.00	Min. :18.00	Min. : 15000	Min. :0.0000
## 1st Qu.:	:0.00	1st Qu.:29.75	1st Qu.: 43000	1st Qu.:0.0000
## Median	:0.00	Median :37.00	Median : 70000	Median :0.0000
## Mean	:0.49	Mean :37.66	Mean : 69742	Mean :0.3575
## 3rd Qu.:	:1.00	3rd Qu.:46.00	3rd Qu.: 88000	3rd Qu.:1.0000
## Max.	:1.00	Max. :60.00	Max. :150000	Max. :1.0000

```
SUV
```

```
## # A tibble: 400 x 4  
##      GEN  AGE  SAL  SUV  
##    <dbl> <dbl> <dbl> <dbl>  
## 1     0   26 43000     0  
## 2     0   27 57000     0  
## 3     0   27 84000     0  
## 4     0   32 150000    1  
## 5     0   35 65000     0  
## 6     0   26 80000     0  
## 7     0   26 52000     0  
## 8     0   48 29000     1  
## 9     0   47 49000     1  
## 10    0   45 22000     1  
## # i 390 more rows
```

#Question 1

#Can you predict whether or not a person purchases a SUV (1 = purchased a SUV, 0 = did not purchase a SUV) based on their age (AGE), annual salary (SAL) and gender (GEN) (Male = 1, Female = 0).

#Part A

#Fit the logistic regression model that regresses SUV on the predictors AGE, SAL and GEN.

```
modell1 <- glm(SUV ~ GEN + SAL + AGE, family = "binomial", data = SUV)
summary(modell1)
```

```
##
## Call:
## glm(formula = SUV ~ GEN + SAL + AGE, family = "binomial", data = SUV)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.278e+01  1.359e+00  -9.405  < 2e-16 ***
## GEN          3.338e-01  3.052e-01   1.094    0.274
## SAL          3.644e-05  5.473e-06   6.659 2.77e-11 ***
## AGE          2.370e-01  2.638e-02   8.984  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 521.57  on 399  degrees of freedom
## Residual deviance: 275.84  on 396  degrees of freedom
## AIC: 283.84
##
## Number of Fisher Scoring iterations: 6
```

```
modell1
```

```
##
## Call:  glm(formula = SUV ~ GEN + SAL + AGE, family = "binomial", data = SUV)
##
## Coefficients:
## (Intercept)          GEN          SAL          AGE
## -1.278e+01    3.338e-01    3.644e-05    2.370e-01
##
## Degrees of Freedom: 399 Total (i.e. Null);  396 Residual
## Null Deviance:      521.6
## Residual Deviance: 275.8    AIC: 283.8
```

$$\text{SUV} = -1.278 + 3.38 \times 10^{-1} \times \text{GENERATION} + 3.644 \times 10^{-5} \times \text{SALARY} + 2.370 \times 10^{-1} \times \text{AGE}$$

#Part B

#Compute the estimate odds ratio for purchasing a SUV for persons who are male and those who are female, controlling for age and salary.

```
exp(3.338*10^-1)
```

```
## [1] 1.396264
```

```
#Part C
```

```
#Interpret your answer in part (B) in the context of the problem.
```

It is the odds ratio of a dude getting a SUV versus a girl getting a SUV. Someone who is Male is 1.4 times more likely to get an SUV than someone who is Female.

```
#Part D
```

```
#Construct a 95% confidence interval for the estimate odds ratio in part (B).
```

```
qnorm(0.975)
```

```
## [1] 1.959964
```

```
exp(3.338*10^-1 + 1.959964 * 3.052*10^-1)
```

```
## [1] 2.539535
```

```
exp(3.338*10^-1 - 1.959964 * 3.052*10^-1)
```

```
## [1] 0.767681
```

```
(0.767681, 2.539535)
```

There is a 95% certainty that the odds ratio between a guy getting an SUV and a girl getting a SUV lies between 0.767681 and 2.539535

```
#Part E
```

```
#Test the null hypothesis that the adjusted odds ratio from part (B) equals 1. Use the Wald test.
```

The null hypothesis is that there is no statistically significant difference between the chance of a guy buying a SUV and a girl buying a SUV.

```
waldtest(model1, test="Chisq", terms = 1)
```

```
## Wald test
```

```
##
```

```
## Model 1: SUV ~ GEN + SAL + AGE
```

```
## Model 2: SUV ~ SAL + AGE
```

```
##   Res.Df Df    Chisq Pr(>Chisq)
```

```
## 1      396
```

```
## 2      397 -1 1.1963      0.2741
```

Since the p-value, 0.2741, exceeds the alpha value of 0.05, we fail to reject the HO!!! Our confidence interval has insufficient evidence to determine that there is a difference in the probability of purchasing an SUV depending on if the buyer is a male or female.

```
#Part F
```

```
#Test the null hypothesis that the adjusted odds ratio from part (B) equals 1. Use the LR test.
```

```
model2 <- glm(SUV ~ SAL + AGE, family = "binomial", data = SUV)
summary(model2)
```

```
##
## Call:
## glm(formula = SUV ~ SAL + AGE, family = "binomial", data = SUV)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.243e+01  1.300e+00  -9.566  < 2e-16 ***
## SAL          3.590e-05  5.429e-06   6.613 3.77e-11 ***
## AGE          2.335e-01  2.591e-02   9.013  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 521.57  on 399  degrees of freedom
## Residual deviance: 277.05  on 397  degrees of freedom
## AIC: 283.05
##
## Number of Fisher Scoring iterations: 6
```

```
model2
```

```
##
## Call:  glm(formula = SUV ~ SAL + AGE, family = "binomial", data = SUV)
##
## Coefficients:
## (Intercept)          SAL          AGE
## -1.243e+01    3.590e-05    2.335e-01
##
## Degrees of Freedom: 399 Total (i.e. Null);  397 Residual
## Null Deviance:      521.6
## Residual Deviance: 277.1    AIC: 283.1
```

```
277.05-275.84
```

```
## [1] 1.21
```

```
1-pchisq(1.21, df=1)
```

```
## [1] 0.2713321
```

Since the p-value, 1.21, exceeds the alpha value, we once again fail to reject the H_0 !!! Our confidence interval has insufficient evidence to determine that there is a difference in the probability of purchasing an SUV depending on if the buyer is a male or female.

#Part G

#Calculate the estimated odds ratio for two people, one with an age of 40 and the other a 52 year old, controlling for salary and gender.

```
exp(12*2.370*10^-1)
```

```
## [1] 17.18437
```

```
#Part H
```

```
#Fit a logistic model that regresses SUV on the predictors AGE, SAL, GEN and the interaction term between AGE and GEN.
```

```
model3 <- glm(SUV ~ GEN + SAL + AGE + AGE:GEN, family = "binomial", data = SUV)
summary(model3)
```

```
##
## Call:
## glm(formula = SUV ~ GEN + SAL + AGE + AGE:GEN, family = "binomial",
##      data = SUV)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.213e+01  1.525e+00 -7.951 1.84e-15 ***
## GEN         -1.214e+00  1.877e+00 -0.647   0.518
## SAL          3.647e-05  5.473e-06  6.664 2.66e-11 ***
## AGE          2.210e-01  3.150e-02  7.017 2.26e-12 ***
## GEN:AGE       3.835e-02  4.593e-02  0.835   0.404
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 521.57  on 399  degrees of freedom
## Residual deviance: 275.14  on 395  degrees of freedom
## AIC: 285.14
##
## Number of Fisher Scoring iterations: 6
```

```
model3
```

```
##
## Call:  glm(formula = SUV ~ GEN + SAL + AGE + AGE:GEN, family = "binomial",
##      data = SUV)
##
## Coefficients:
## (Intercept)          GEN          SAL          AGE          GEN:AGE
## -1.213e+01   -1.214e+00    3.647e-05    2.210e-01    3.835e-02
##
## Degrees of Freedom: 399 Total (i.e. Null);  395 Residual
## Null Deviance:      521.6
## Residual Deviance: 275.1    AIC: 285.1
```

```
SUV = -12.13 - 1.214 * Generation + 3.657 * 10 ^ -5 * Salary + 2.21 * 10 ^ -1 * Age + 3.835 * 10 ^ -2 *
Gen:Age
```

```
#Part I
```

#Using the model from (H), compute the 95% confidence interval for the estimated adjusted odds ratio for the effect of gender, controlling for AGE and SAL when a person is 25 years old.

```
round(vcov(model3), 3)
```

```
##           (Intercept)      GEN SAL      AGE GEN:AGE
## (Intercept)      2.327 -1.402   0 -0.046   0.032
## GEN             -1.402  3.524   0  0.033  -0.085
## SAL              0.000  0.000   0  0.000   0.000
## AGE             -0.046  0.033   0  0.001  -0.001
## GEN:AGE          0.032 -0.085   0 -0.001   0.002
```

```
1.877^2 + 25^2*(4.593 * 10 ^ -2) + 2 * 1.877 * (-0.085)
```

```
## [1] 31.91029
```

```
exp(-1.214 + 25*3.84*10^-2 + 1.96 * (1.877^2 + 25^2 * 0.0459^2 + 2*25*(-0.0851)))
```

```
## [1] 2.44092
```

```
exp(-1.214 + 25*3.84*10^-2 - 1.96 * (1.877^2 + 25^2 * 0.0459^2 + 2*25*(-0.0851)))
```

```
## [1] 0.2465045
```

We are 95% confident that the actual odds ratio lies between 0.2465045 and 2.44092.

#Part J

#What does the confidence interval in part (I) tell you?

There is a 95% certainty that the odds ratio lies within the interval. Since the interval includes 1, it means that there is a chance that there is no difference in SUV purchasing between genders when all other variables are controlled for.